Issue	**1**
Factory ID	**25130**
Date of Issue	**27-May-2026**

Interloop Ltd. (HD05) co-located w/Interloop Ltd. Apparel Park

Bypass Khurrianwala Jaranwala Road,Khurrianwala,Tehsil Jaranwala,District Faisalabad
38000 Faisalabad
Punjab Pakistan

Latitude: [31.452469], longitude: [73.281943]
Date of inspection: 06-Apr-2026

Electrical Safety Inspection Report

Observations & Actions



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Reviewed by	**Shahzeb Anis**
Approved by	**Colm Quinn**

Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 06-04-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 06-Apr-2026, Azka Riaz and Muhammad Bilal Butt on behalf of the Pakistan Accord carried out an initial electrical inspection for Interloop Ltd. (HD05) co-located w/Interloop Ltd. Apparel Park, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Naveed-Ur-Rehman (DGM EHS), Mr. Muhammad Ghayas (Manager EHS), Mr. Muhammad Abbas (Sr. Manager Electrical), Mr. Zaheer Ahmad (Civil Engineer) along with other staff members of the factory. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were also present, and the team was led by Muhammad Bilal Butt, Pakistan Accord.

The factory operates in a total of five shifts. Approximately 7000 workers operate during a single general shift at peak occupancy. The factory employs 13000 workers, with 1600 workers being female.

The factory comprises 26 buildings and 33 ancillary structures owned by Interloop Apparel Park. The main buildings and ancillary structures are further divided into specific areas, namely Hosiery Division 5, Hosiery Division 6, Yarn Dyeing Area, Apparel Plant 2, Common Area, and Club & Hostel Area, the details of which are provided below:

Buildings:

Hosiery Division 5:

1. **Office Block Building** is a 2 storey G+1. The building is used for unoccupied (roof); offices and meeting room (first floor); offices, meeting room and kitchen (ground floor). The building was constructed in 2021.
2. **Canteen Building** is a 2 storey G+1. The building is used for unoccupied (roof); dining area (first floor); dining (differently abled), prayer area-male, prayer area- female and kitchen (ground floor). The building was constructed in 2021.
3. **Processing lab Building** is a ground only. The building is used for unoccupied (roof); accessory stores, sampling area, waste management area, processing area, labs, offices and chemical storage (ground floor). The building was constructed in 2021.
4. V**alue stream unit 2 Building – HD 5** is a ground only. The building is used for unoccupied (roof); electronic lab, control room, AHU, LT room, store, offices, knitting, finishing, metal free zone finish good store, accessory store (ground floor). The building was constructed in 2021.
5. **Value stream unit 1 Building – HD 5** is a ground only. The building is used for unoccupied (roof); electronic lab, red tag 1 area, control room, AHU, LT room, store, offices, knitting, finishing, metal free zone finish good store, spare store, red tag 2 area, charging point , chemical store (ground floor). The building was constructed in 2021.

Hosiery Division 6:

1. **Value stream unit 1 Building – HD 6** is a ground only. The building is used for unoccupied (roof); AHU, LT room, store, finishing, offices, electronic lab, finish good store, office, prayer area, sampling and accessory store (ground floor). The building was constructed in 2025.
2. **Value stream unit 2 Building – HD 6** is a ground only. The building is used for unoccupied (roof); AHU, LT room, store, finishing, offices, electronic

lab, finish good store, office, prayer area, sampling, dining, kitchen and store (ground floor). The building was constructed in 2025.

Yarn Dying Area:

1. **Yarn Dying Building** is a ground only. The building is used for unoccupied (roof); lab, office and under construction area dying machines (mezzanine); yarn dying, store, AHU, prayer area, LT room and RO Plant (ground floor). The building was constructed in 2025.
2. **Yarn Dying- Chemical Store Building** is a ground only. The building is used for unoccupied (roof); yarn dying store and workshop (ground floor). The building was constructed in 2025.

Apparel Plant 2:

1. **Garments Block Building** is a 3 storey G+2. The building is used for unoccupied (roof); stitching, dining, kitchen, mosques, storage & offices (second floor); stitching, dining, kitchen, mosques, storage & offices (first floor); cutting, store, embroidery and offices (ground floor). The building was constructed in 2023.

2. Dye House Building is a ground only G+M. The building is used for unoccupied (roof); lab and offices (mezzanine); dying, finishing, washing, chemical, dispensing and storage, LT room (1), LT room (2) and Thermal oil heaters (ground floor). The building was constructed in 2023.

3. **Utility Area Apparel Building** is a 2 storey G+1. The building is used for unoccupied (roof); offices (first floor); chiller & compressor area, electric panels, RO Plant, electrical workshop, mechanical workshop 1, mechanical workshop 2, waste store, leftover store, HVAC workshop (ground floor). The building was constructed in 2023.
4. **Marketing Building is a 2 storey G+1.** The building is used for unoccupied (roof); sampling, embroidery and offices (first floor); under-construction (ground floor). The building was constructed in 2025.

Common area:

1. **Technical Training School Building** is a 4 storey G+3. The building is used for unoccupied (roof); under construction (third floor); training halls, dining area, rest area, offices and storage (second floor); training halls, dining area, rest area, offices and storage (first floor); workshop 1, workshop 2, store, workshop 3 (ground floor). The building was constructed in 2021.
2. **Workshop Building is a 2 storey** G+1. The building is used for unoccupied (roof); offices (first floor); under-construction (ground floor). The building was constructed in 2021.
3. **Main Gate Office Building** is a 2 storey G+1. The building is used for unoccupied (roof); offices and under construction area (first floor); control room, HR & requirement Offices, time office, security offices, driver sitting , rest area and TUC shop (ground floor). The building was constructed in 2021.

Club & Hostel area:

1. **Central Block 1 & Central Block 2 Buildings** are typical 3 storey G+2. The building is used for solar (roof); dormitories (second & first floor); dormitories, kitchen and dining (ground floor). The building was constructed in 2025.
2. **Outer block 1, Outer block 2, Outer block 3, Outer block 4 & Outer block 5 Buildings** are typical 3 storey G+2. The building is used for solar (roof); dormitories (second, first and ground floor). The building was constructed in 2025.
3. **Executive Hostel (Male) & Executive Hostel (Female) Buildings** are typical 3 storey G+2. The building is used for unoccupied (roof); dormitories (second, first and ground floor). The building was constructed in 2025.
4. **Club Building** is a 2 storey G+1. The building is used for unoccupied (roof); gym and rest room (first floor); dining and sitting (ground floor). The building was constructed in 2025.

Ancillaries structures:

Hosiery Division 5:

1. **Utility 1** is a ground only. The building is used for unoccupied (roof); chillers, compressors and workshops (ground floor). The building was constructed in 2021.
2. **Utility 2** is a ground only. The building is used for unoccupied (roof); RO plant and workshop (ground floor). The building was constructed in 2021.
3. **Engineering offices** is a ground only. The building is used for unoccupied (roof); offices (ground floor). The building was constructed in 2021.
4. **Storage** is a ground only. The building is used for unoccupied (roof); storage (ground floor). The building was constructed in 2021.
5. **Carpenter & Fire boost workshop** is a ground only. The building is used for unoccupied (roof); workshop and fire equipment store (ground floor). The building was constructed in 2021.

Hosiery Division 6:

1. **Time office** is a ground only. The building is used for unoccupied (roof); offices (ground floor). The building was constructed in 2025.

Apparel Plant 2:

1. **Wastewater treatment plant** is a ground only. The building is used for unoccupied (roof); MBR, Lab, electric panels and blower area (ground floor). The building was constructed in 2023.
2. **Chemical store apparel** is a ground only. The building is used for unoccupied (roof); chemical store, scrap store and hazardous material store (ground floor). The building was constructed in 2023.

Common area:

1. **Apparel gate office** is a ground only. The building is used for unoccupied (roof); offices (ground floor). The building was constructed in 2023.
2. **Fair price shop** is a ground only. The building is used for unoccupied (roof); shop and ATMs (ground floor). The building was constructed in 2021.
3. **Dispensary** is a ground only. The building is used for unoccupied (roof); medical health unit (ground floor). The building was constructed in 2021.
4. **Storage/ Locker room** is a ground only. The building is used for unoccupied (roof); storage (ground floor). The building was constructed in 2024.
5. **Masjid** is a ground only. The building is used for unoccupied (roof); prayer area (ground floor). The building was constructed in 2021.
6. **Daycare** is a ground only. The building is used for unoccupied (roof); daycare activities and store (ground floor). The building was constructed in 2021.
7. **LT room and OHWT** is a ground-only structure. The building is used for water storage (roof); LT panels (ground floor). The building was constructed in 2021.
8. **Hydrant room** is a basement only. The building is used for unoccupied (roof); pumping system (ground floor). The building was constructed in 2021.
9. **Hosiery gate office** is a ground only. The building is used for unoccupied (roof); offices (ground floor). The building was constructed in 2021.

Club & Hostel area:

1. **Colony gate office** is a ground only. The building is used for unoccupied (roof); offices (ground floor). The building was constructed in 2023.
2. **Security office & ATM** is a ground only. The building is used for unoccupied (roof); offices and ATM (ground floor). The building was constructed in 2024.
3. **Colony workshops** is a ground only structure. The building is used for unoccupied (roof); workshops and LT room (ground floor). The building was

constructed in 2024.

4. **Drivers room** is a ground only. The building is used for unoccupied (roof); waiting area (ground floor). The building was constructed in 2024.
5. **Colony OHWT** is a ground only building. The building is used for water storage (roof); RO Plant (ground floor). The building was constructed in 2024.
6. **Colony Masjid** is a ground only building. The building is used for water storage (roof); prayer area (ground floor). The building was constructed in 2024.
7. **Cafeteria and squash** is a ground only. The building is used for water storage (roof); canteen, dining and squash court (ground floor). The building was constructed in 2026.
8. **Apartments** is a 2 storey G+1 building. The building is used for water storage (roof); sleeping area (first floor & ground floor).The building was constructed in 2026.

Utilities:

1. **Parking shed** is a ground only. The building is used for solar (roof); car parking (ground floor). The building was constructed in 2021.
2. **Boiler gate office** is a ground only structure. The building is used for unoccupied (roof); offices (ground floor). The building was constructed in 2023.
3. **Generator room 1** is a ground only structure. The building is used for unoccupied (roof); generators (ground floor). The building was constructed in 2023.
4. **Generator room 2** is a ground only. The building is used for unoccupied (roof); generators (ground floor). The building was constructed in 2023.
5. **Transformer room** is a ground only structure. The building is used for unoccupied (roof); transformers (ground floor). The building was constructed in 2023.
6. **LT room** is a ground only structure. The building is used for unoccupied (roof); LT panels (ground floor). The building was constructed in 2023.
7. **Gas boiler area** is a ground only Shed. The building is used for unoccupied (roof); gas boilers and control room (ground floor). The building was constructed in 2023.
8. **Biomass boiler area** is a 2 storey G+1 Shed. The building is used for unoccupied (roof); control room (first floor), boiler and electric panels (ground floor). The building was constructed in 2023.
9. **Storage shed area** is a ground only Shed. The building is used for unoccupied (roof); biomass storage (ground floor). The building was constructed in 2023.

A summary of the main concerns in relation to life safety systems identified during the inspection (non-exhaustive) that should be actioned is as follows:

• **Abnormal Temperature:** Within electrical panels, abnormal temperatures were recorded at various locations (refer to section 7.2 for further details). Hotspots within panels can cause short circuits and fire due to insulation damage and overheating of cables or electrical components.
• **Fire Alarm Detection System:** Multiple addressable fire alarm systems were installed across the facility; however, deficiencies were observed in detector coverage, placement, system integration, and some panels showed fault indications, along with missing certification documents. Inadequate detection and lack of proper interfacing may delay fire response, hinder automatic controls, and increase risks to life safety and property.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents

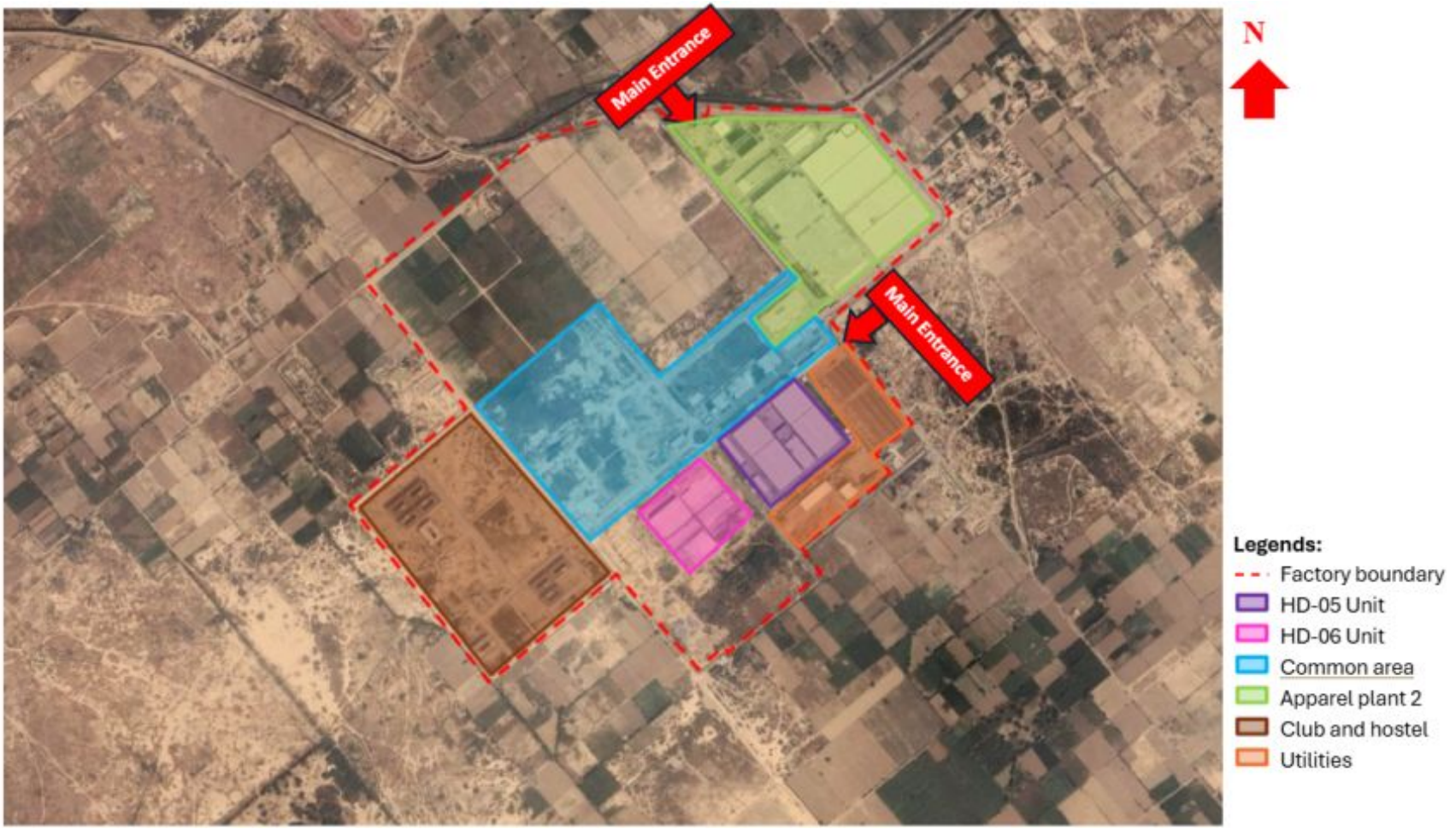


Figure 1.1. Satellite view of the premises. Red dashed line indicates site boundary and highlighted area indicates buildings inspected

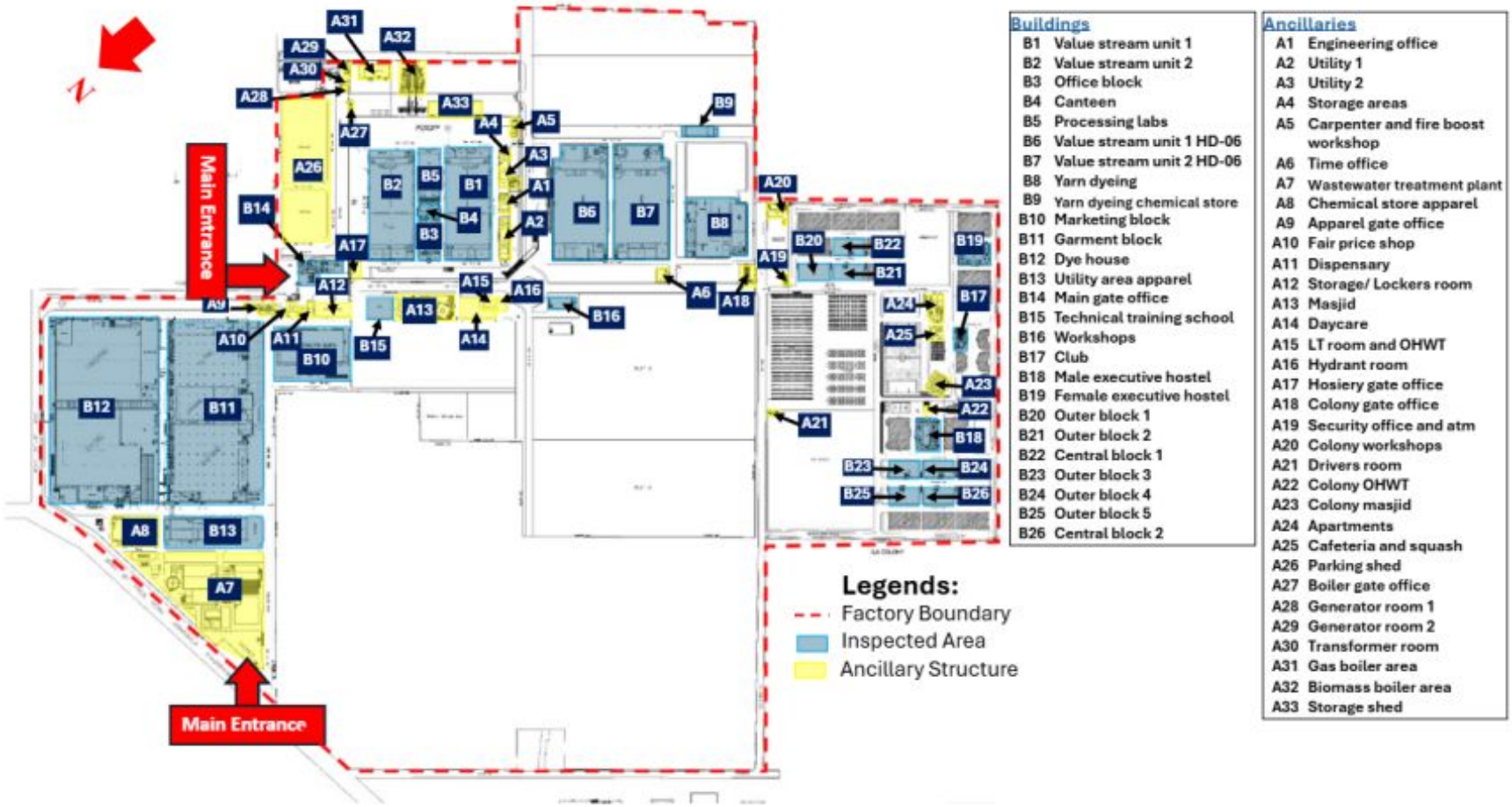


Figure 1.2. Site layout plan - Red dashed line indicates site boundary and highlighted area indicates buildings inspected

2.1 Building Use

The factory comprises twenty-six main buildings and thirty-three ancillary structures. The floor-wise activity details are as follows:

Main Buildings:

Hosiery Division 5

Office Block Building:
L00 – Ground Floor: Offices, meeting room, and kitchen.
L01 – First Floor: Offices and meeting room.
L99 – Roof: Unoccupied.
Canteen Building:
L00 – Ground Floor: Dining (differently abled), prayer area (male & female), and kitchen.
L01 – First Floor: Dining area.
L99 – Roof: Unoccupied.
Processing Lab Building:
L00 – Ground Floor: Accessory stores, sampling, waste management, processing, labs, offices, and chemical storage.
L99 – Roof: Unoccupied.
Value Stream Unit 2 – HD 5:
L00 – Ground Floor: Electronic lab, control room, AHU, LT room, store, offices, knitting, finishing, metal-free finished goods store, and accessory store.
L99 – Roof: Unoccupied.
Value Stream Unit 1 – HD 5:
L00 – Ground Floor: Electronic lab, red tag areas (1 & 2), control room, AHU, LT room, store, offices, knitting, finishing, metal-free finished goods store, spare store, charging point, and chemical store.
L99 – Roof: Unoccupied.

Hosiery Division 6

Value Stream Unit 1 – HD 6:
L00 – Ground Floor: AHU, LT room, store, finishing, offices, electronic lab, finished goods store, prayer area, sampling, and accessory store.
L99 – Roof: Unoccupied.
Value Stream Unit 2 – HD 6:
L00 – Ground Floor: AHU, LT room, store, finishing, offices, electronic lab, finished goods store, prayer area, sampling, dining, kitchen, and store.

L99 – Roof: Unoccupied.

Yarn Dyeing Area

Yarn Dyeing Building:
L00 – Ground Floor: Yarn dyeing, store, AHU, prayer area, LT room, and RO plant.
M01 – Mezzanine Floor: Lab, office, and dyeing machines (under construction).
L99 – Roof: Unoccupied.
Yarn Dyeing – Chemical Store Building:
L00 – Ground Floor: Yarn dyeing store and workshop.
L99 – Roof: Unoccupied.

Apparel Plant 2

Garments Block Building:
L00 – Ground Floor: Cutting, store, embroidery, and offices.
L01 – First Floor: Stitching, dining, kitchen, mosques, storage, and offices.
L02 – Second Floor: Stitching, dining, kitchen, mosques, storage, and offices.
L99 – Roof: Unoccupied.
Dye House Building:
L00 – Ground Floor: Dyeing, finishing, washing, chemical dispensing, storage, LT rooms (1 & 2), and thermal oil heaters.
M01 – Mezzanine Floor: Lab and offices.
L99 – Roof: Unoccupied.
Utility Area Apparel Building:
L00 – Ground Floor: Chillers, compressors, electric panels, RO plant, electrical and mechanical workshops, waste store, leftover store, and HVAC workshop.
L01 – First Floor: Offices.
L99 – Roof: Unoccupied.
Marketing Building:
L00 – Ground Floor: Under construction.
L01 – First Floor: Sampling, embroidery, and offices.
L99 – Roof: Unoccupied.

Common Area

Technical Training School Building:
L00 – Ground Floor: Workshops and store.

L01 – First Floor: Training halls, dining, rest area, offices, and storage.
L02 – Second Floor: Training halls, dining, rest area, offices, and storage.
L03 – Third Floor: Under construction.
L99 – Roof: Unoccupied.

Workshop Building:

L00 – Ground Floor: Under construction.
L01 – First Floor: Offices.
L99 – Roof: Unoccupied.

Main Gate Office Building:

L00 – Ground Floor: Control room, HR & recruitment, time office, security offices, driver sitting, rest area, and TUC shop.
L01 – First Floor: Offices and under construction area.
L99 – Roof: Unoccupied.

Club & Hostel Area

Central Blocks 1 & 2:

L00 – Ground Floor: Dormitories, kitchen, and dining.
L01 – First Floor: Dormitories.
L02 – Second Floor: Dormitories.
L99 – Roof: Solar.

Outer Blocks 1–5:

L00 – Ground Floor: Dormitories.
L01 – First Floor: Dormitories.
L02 – Second Floor: Dormitories.
L99 – Roof: Solar.

Executive Hostels (Male & Female):

L00 – Ground Floor: Dormitories.
L01 – First Floor: Dormitories.
L02 – Second Floor: Dormitories.
L99 – Roof: Unoccupied.

Club Building:

L00 – Ground Floor: Dining and sitting.
L01 – First Floor: Gym and rest room.
L99 – Roof: Unoccupied.

Ancillary Structures:

Utility 1 (HD 5):
L00 – Ground Floor: Chillers, compressors, and workshops.
L99 – Roof: Unoccupied.
Utility 2 (HD 5):
L00 – Ground Floor: RO plant and workshop.
L99 – Roof: Unoccupied.
Engineering Offices (HD 5):
L00 – Ground Floor: Offices.
L99 – Roof: Unoccupied.
Storage (HD 5):
L00 – Ground Floor: Storage.
L99 – Roof: Unoccupied.
Workshops (HD 5):
L00 – Ground Floor: Workshop and fire equipment store.
L99 – Roof: Unoccupied.
Wastewater treatment plant (Apparel):
L00 – Ground Floor: MBR, lab, electric panels, and blower area.
L99 – Roof: Unoccupied.
Chemical Store (Apparel):
L00 – Ground Floor: Chemical, scrap, and hazardous material storage.
L99 – Roof: Unoccupied.
Apparel gate office (Common Area):
L00 – Ground Floor: Offices.
L99 – Roof: Unoccupied.
Fair price shop (Common Area):
L00 – Ground Floor: Shop and ATMs.
L99 – Roof: Unoccupied.
Dispensary (Common Area):
L00 – Ground Floor: Medical health unit.
L99 – Roof: Unoccupied.
Storage/ Locker room (Common Area):

L00 – Ground Floor: Storage.
L99 – Roof: Unoccupied.

Masjid (Common Area):
L00 – Ground Floor: Prayer area.
L99 – Roof: Unoccupied.

Daycare (Common Area):
L00 – Ground Floor: Daycare activities and store.
L99 – Roof: Unoccupied.

LT room and OHWT (Common Area):
L00 – Ground Floor: LT panels.
L99 – Roof: Water storage.

Hydrant room (Common Area):
L00 – Ground Floor: Pumping system.
L99 – Roof: Unoccupied.

Hosiery gate office (Common Area):
L00 – Ground Floor: Offices.
L99 – Roof: Unoccupied.

Parking Shed (Utilities):
L00 – Ground Floor: Car parking.
L99 – Roof: Solar.

Boiler Gate Office (Utilities):
L00 – Ground Floor: Offices.
L99 – Roof: Unoccupied.

Generator Room 1 (Utilities):
L00 – Ground Floor: Generators.
L99 – Roof: Unoccupied.

Generator Room 2 (Utilities):
L00 – Ground Floor: Generators.
L99 – Roof: Unoccupied.

Transformer Room (Utilities):
L00 – Ground Floor: Transformers.
L99 – Roof: Unoccupied.

LT Room (Utilities):
L00 – Ground Floor: LT panels.
L99 – Roof: Unoccupied.

Gas Boiler Area (Utilities):

L00 – Ground Floor: Gas boilers and control room.

L99 – Roof: Unoccupied.

Biomass Boiler Area (Utilities):

L00 – Ground Floor: Boiler and electric panels.

L01 – First Floor: Control room.

L99 – Roof: Unoccupied.

Storage Shed Area (Utilities):

L00 – Ground Floor: Biomass storage.

L99 – Roof: Unoccupied.

2.2 Utility Source

The facility is supplied with five 11 kV connections from Faisalabad Electric Supply Company (FESCO) at the Interloop Energy Division as its primary power source. Of these, two feeders supply the Apparel Plant, one supplies Hosiery Division 5, one supplies Hosiery Division 6, and one is designated for emergency use. The incoming power is stepped down in substations at respective locations and distributed across the facility through underground cables to various distribution boards. In addition, a 2.5 MW solar power system is synchronized with the main utility supply.

2.3 Backup source

The backup power supply is provided by the Interloop Power House (not included within the scope of this inspection). Additionally, two diesel generators, rated at 1250 kVA and 550 kVA and located in the Boiler House, are available as emergency backup sources. The fire alarm and emergency lighting systems are also equipped with built-in battery backup provisions.

3. Documentation

This table shows the details of the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No	-	Electrical and power layouts are not provided.
2	Single Line Diagrams	Yes	23-02-2026	Missing essential details.
3	Fire Alarm Drawing	Yes	Not mentioned	Incomplete and missing essential details.
4	Earth Test Report	Yes	08-10-2025	Not provided for all earthing pits, specifically HD5 & HD6.
5	Thermographic Test Report	No	-	-
6	Insulation Resistance Test Report	Yes	Not mentioned	Not provided for all HT & LT cables.
7	LPS Risk Assessments	No	-	-
8	Periodic Maintenance Record	Yes	Feb-2026	Emergency lighting and fire alarm test reports provided.
9	Transformer Oil Testing Report	Yes	09-Feb-2025	Not provided for oil transformers located in the boiler house.

4. Electrical Safety Measures

During our electrical site inspection of Interloop Ltd. (HD05) co-located w/Interloop Ltd. Apparel Park, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Interloop Ltd. (HD05) co-located w/Interloop Ltd. Apparel Park. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202604-065560

Timeline	Remediation Plan
Medium Term CAP-0077708	Factories should begin remediation of CAP items, such as cleaning, cable organization, and basic corrections, using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202604-065008

Timeline	Remediation Plan
Medium Term CAP-0077047	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077048	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the submitted electrical single-line diagrams (SLDs) lacked master layouts for the overall power distribution system and were missing several essential details, such as circuit tags, cable sizes, and short-circuit ratings in most SLDs. Furthermore, the SLDs did not match the current site installations.

Single-line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202604-065006

Timeline	Remediation Plan
Medium Term CAP-0077063	Update and develop comprehensive electrical single-line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077064	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202604-065007

Timeline	Remediation Plan
Medium Term CAP-0077053	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077054	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey, the circuit breakers installed throughout the facility, inside the electrical panels and distribution boards, were inspected.

It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes. Specifically noted at the following locations:

a. In the LT panel of Hosiery Division 5, a 250A gate office circuit breaker, adjusted at 175A, was connected to 16 mm² PVC cables.
b. In the LT panel of Hosiery Division 5, a 400A ETP circuit breaker was connected to 70 mm² PVC cables.
c. In the LT panel of Boiler House, a 400A ETP circuit breaker was connected to 70 mm² PVC cables.
d. In the LT panel of Apparel Dye House, a 400A circuit breaker was connected to 25 mm² PVC cables.
e. In the Mess distribution board of Executive Female Hostel, a 160A circuit breaker was connected to 10 mm² PVC cables.
f. In the Booster pump distribution board of Colony OHWT, a 100A circuit breaker was connected to 6 mm² PVC cables.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202604-064998

Timeline	Remediation Plan
Medium Term CAP-0077682	Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - 160A circuit breaker connected to 10 mm² PVC cables, mess distribution board, Executive Female Hostel

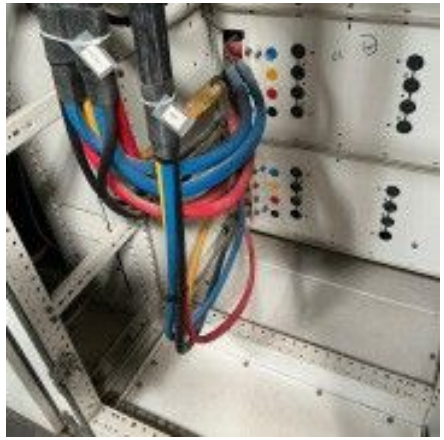


Figure 2.3 - 400 A circuit breaker connected to 70 mm² PVC cables, LT panel of Hosiery Division 5



Figure 2.2 - 100 A circuit breaker connected to 6 mm² PVC cables, booster pump distribution board, Colony OHWT



Figure 2.4 - 400 A circuit breaker connected to 70 mm² PVC cables, in gas boiler LT panel, Boiler House

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:
i. 119°C, 110°C – LT2 Unit 2, Hosiery Division 5.
ii. 117°C – Module DB 20, Second Floor, Cut to Pack Building, Garments Block.
iii. 116°C, 113°C – Cable connection points of MCC breakers, and 97.6°C, 81.5°C – Cable connector terminals at exhaust fan in Finishing Area, Apparel Dye House.
iv. 112°C – Exhaust Fan 3 Magnetic Switch, Mess Kitchen Unit 2, Value Stream Unit 2.
v. 103°C – Dry Type Transformer, LT1, Apparel Plant.
vi. 97.7°C, 86.1°C, 85.2°C – 800 kVAR PFI Panel LT1, Apparel Plant (Sections 1 & 2).
vii. 93.2°C – 3500 kVA Dry Type Transformer, LT Room, Yarn Dyeing.
viii. 87.3°C – LT1 HD-06, 2500 kVA Dry Type Transformer, Apparel Plant.
ix. 83.4°C, 81.2°C – Cooling Fan 5 and Cooling Water Pump 2, Compressor & Chiller Control Room, Engineering Block, Apparel Plant.
x. 82.2°C – B1-MCC-5, Biomass Boiler 1 Control Room, Boiler House.
xi. 77.1°C, 73.5°C – B2-MCC-5, Biomass Control Room, Boiler House.

xii. 78.9°C – LT01, HD-05 Exhaust Fan Switch.
xiii. 71.2°C – Baseplate LT Panel, Solar Boiler House.
xiv. 70°C – Magnetic Contactor LT3, Hosiery Division 5

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC-insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment. F-202604-064999

Timeline	Remediation Plan
Immediate CAP-0077572	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0077573	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature, as illustrated in the example images below. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0077574	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit the thermographic test report to the Pakistan Accord for review.

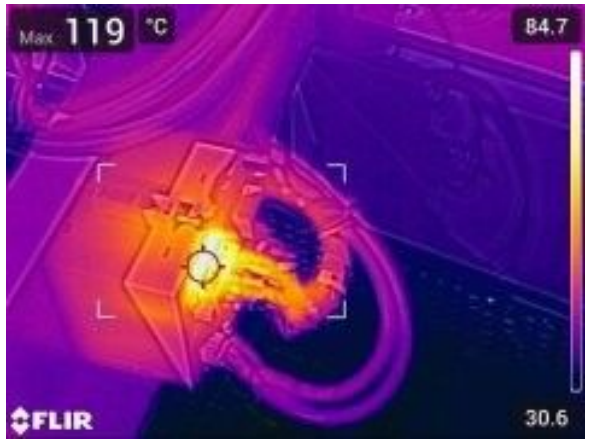


Figure 2.5 - Hot spot found in LT panel Unit 2, Hosiery Division 5



Figure 2.7 - Hot spot found on Module DB 20, Second Floor, Cut to Pack Building, Garments Block



Figure 2.6 - Hot spot found in LT panel Unit 2, Hosiery Division 5



Figure 2.8 - Hot spot found on Module DB 20, Second Floor, Cut to Pack Building, Garments Block

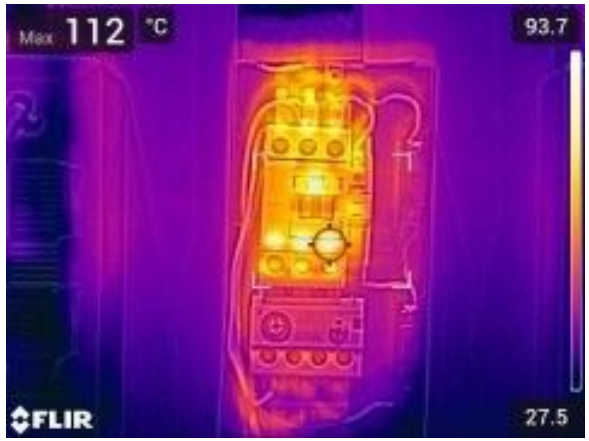


Figure 2.9 - Hot spot found on Exhaust Fan 3 Magnetic Switch, Mess Kitchen Unit 2, Value Stream Unit 2



Figure 2.10 - Hot spot found on Exhaust Fan 3 Magnetic Switch, Mess Kitchen Unit 2, Value Stream Unit 2

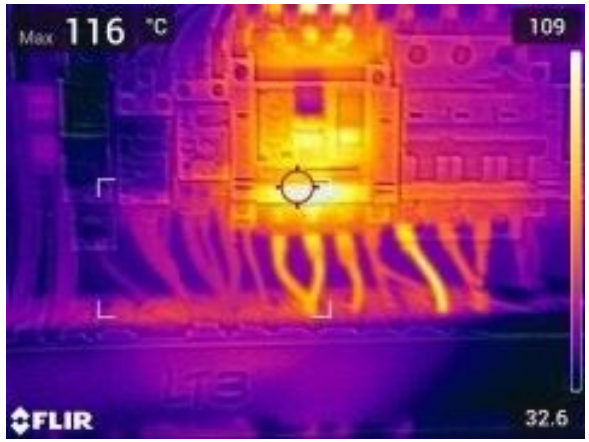


Figure 2.11 - Hotspot found on MCC breaker in finishing distribution board, Apparel Dye House.



Figure 2.12 - Hotspot found on MCC breaker in finishing distribution board, Apparel Dye House.

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. No earthing system design documentation was provided by the factory, indicating a lack of verification of the earthing system’s compliance and ongoing performance in accordance with relevant standards.

ii. It was noted that the earth bar was incorrectly connected to the neutral bar, indicating improper grounding practices. This issue was observed specifically in the RO plant distribution board of Colony OHWT, lighting distribution board # 25 of OHWT LT room, the lighting distribution board of Gas Boiler Hall, and the power distribution board #1 on the ground floor of Male Non-Executive Hostel.

iii. Equipotential bonding for the cable trays and metallic panel covers was missing. This issue was observed specifically in the shredder machine distribution board in the waste store area, power distribution board # 1 in Gas Boiler Hall, and the electronics lab distribution board and cable trays in the Garments Block.

iv. The earth pit testing and maintenance records were not provided for all earthing pits, specifically those installed in Hosiery Division 5 & 6.

The lack of a proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202604-065001

Timeline	Remediation Plan
Medium Term CAP-0077664	i. Design and evaluate the facility's earthing system, and develop layout drawings in accordance with BS 7671 18th Edition + A2 (2022). Submit the earthing and neutral layout drawings to the Pakistan Accord for review.
Medium Term CAP-0077665	ii. Disconnect the earth bars from the neutral bars with the distribution boards and ensure proper separation and connection.
Medium Term CAP-0077666	iii. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.

Medium Term CAP-0077667	iv. Implement periodic earth testing per BS 7671 Regulation 643.7.1 to verify grounding effectiveness and submit testing record to Pakistan Accord for review.
Longer Term CAP-0077668	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.13 - Equipotential bonding missing for the panel door in shredder machine distribution board in waste store area



Figure 2.15 - Neutral and earth cables combined in the power distribution board #1 of Male Non-Executive Hostel



Figure 2.14 - Equipotential bonding missing for the cable trays in Garments Block



Figure 2.16 - Neutral and earth cables combined in the RO plant distribution board of Colony OHWT

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables, and busbars throughout the facility.

The booster pump distribution board installed at the treated water storage area and the flood lights distribution board installed at Male Non-Executive Hostel exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements.

The utilization of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202604-064997

Timeline	Remediation Plan
Medium Term CAP-0077697	Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Longer Term CAP-0077698	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.17 - Corrosion observed on booster pump distribution board installed at the treated water storage area



Figure 2.18 - Corrosion observed on flood lights distribution board installed at Male Non-Executive Hostel

7.5 Loop Connection

This relates to the entire facility.

During the inspection, loop connections were identified within the electrical panels. This issue was observed specifically in the ground-floor distribution board of the Technical Training School and in the booster pump distribution board of Hosiery Division 5 Storage area.

In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202604-065000

Timeline	Remediation Plan
Medium Term CAP-0077084	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.19 - Loop connections found in the booster pump distribution board, Hosiery Division 5 Storage area

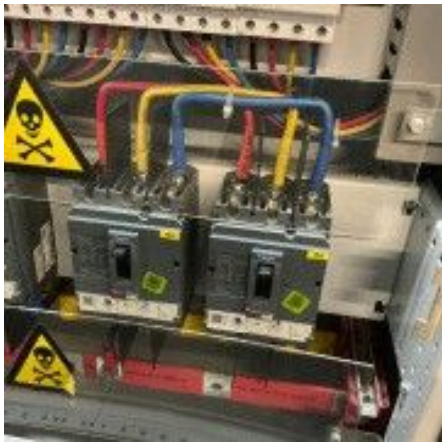


Figure 2.20 - Loop connections found in the ground-floor distribution board, Technical Training School

7.6 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At numerous locations, it was found that the electrical panels do not possess the required space for inspection or easy access to the panels and their components. This issue was observed specifically for the distribution boards in the RO plant area of Hosiery Division 5 Utility 2, on the second floor of Garments Block, and near the Apparel gate office.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202604-065003

Timeline	Remediation Plan
Medium Term CAP-0077083	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.21 - Inadequate accessibility for the distribution board, on the second floor, Garments Block



Figure 2.22 - Inadequate accessibility for the distribution board near Apparel gate office



Figure 2.23 - Inadequate accessibility for the distribution board, in the RO plant area, Hosiery Division 5 Utility 2

7.7 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

A lightning protection system was provided; however, no risk assessment report was provided in accordance with BS EN 62305. The submitted lightning protection design was based on NF C 17-102, which utilizes the Early Streamer Emission (ESE) method. As per BS EN 62305, ESE-based designs are not recognized and therefore do not demonstrate compliance with the standard.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202604-065004

Timeline	Remediation Plan
Medium Term CAP-0077081	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment according to BS EN 62305 (Protection against Lightning) standard. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0077082	If the risk assessment concludes that an LPS is necessary, then hire a qualified engineer to perform an adequacy study of the existing LPS. This study should evaluate whether the current system meets the necessary design and performance standards according to BS EN 62305. If the study finds that the current LPS is inadequate, make the necessary upgrades to bring the system up to standard.



Figure 2.24 - Lightning protection system installed on the rooftop of building 13



Figure 2.25 - Lightning protection system installed on the rooftop of building 27



Figure 2.26 - Lightning protection system installed on the rooftop of building 31

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

Multiple addressable fire alarm control panels are installed to cover the entire facility. However, in certain areas, the provision of manual call points and fire detection was found to be inadequate, the placement of detectors was inappropriate, and non-fire-rated cables were used. Additionally, fire alarm interfaces for the Fire Alarm Detection System (FADS) to control HVAC systems (including dampers), elevators, and electronic access control systems were not provided. Some fire alarm control panels were also observed showing fault indications. Furthermore, the addressable panels and devices lack the necessary documentation and certification to verify their authenticity as listed products.

An incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202604-065010

Timeline	Remediation Plan
Medium Term CAP-0077575	i. Submit design documentation for the new replacement of FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0077576	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.
Longer Term CAP-0077577	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0077578	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.

Longer Term CAP-0077579	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0077580	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.27 - Fire alarm system interfacing not provided with the passenger lifts in Garments Block



Figure 2.28 - Addressable fire alarm control panel showing fault indications installed in the Hosiery Division 5



Figure 2.29 - Fire alarm detection not provided in the solar inverter area, Male Non-executive Hostel



Figure 2.30 - Insufficient fire detection coverage and manual call points in cooking area on the first floor of Garments Block



Figure 2.31 - Addressable fire alarm control panel showing fault indications installed in the Boiler House



Figure 2.32 - Manual call point missing in the fire post room



Figure 2.33 - Manual call point missing in the compressor room of Bio Gas Boiler



Figure 2.34 - Smoke detector found obstructed in the engineering block, Apparel Utility Area



Figure 2.35 - Multidetector installed at excessive height in Hosiery Division 5 Processing

9. Emergency Lighting

9.1 Provided but insufficient

The section refers to the entire facility.

During the inspection, the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency lighting and exit directional signs coverage. This issue was observed in the knitting area spare store of Value Stream Unit-1, the backside of LT panels in Hosiery Division 5 sub-station room, the internal stairs of Garments Block, the internal stairs of Male Non-Executive Hostels, the RO plant room of Colony OHWT ancillary, and the Carpenter & Fire Post ancillary. Additionally, the emergency lighting layouts were missing. The absence of emergency lighting layouts throughout the facility raises concerns regarding the adequacy of illumination provided during power failures or emergencies.

ii. The installed emergency light was found too dim to provide adequate illumination in the Gas Boiler Hall ancillary.

iii. Emergency light and exit sign were found obstructed in the knitting hall of Value Stream Unit-1.

iv. External emergency lighting was not provided, preventing people from reaching a place of safety outside the facility during nighttime evacuations. This issue was observed specifically in the external emergency stairs of the Hosiery Division-5 Canteen building, the Biomass Boiler control room, and the Non-Executive Female Hostel buildings.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202604-065005

Timeline	Remediation Plan

Medium Term CAP-0077123	i. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Prepare and submit the emergency light design and layouts to the Pakistan Accord for review. Install additional emergency luminaires and exit directional signs in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.
Medium Term CAP-0077124	ii. Replace dim emergency lights with higher intensity luminaires that meet the minimum illumination levels specified in the Building Code of Pakistan Fire Safety Provisions 2016. Ensure that all emergency lights provide adequate brightness to facilitate safe evacuation during emergencies.
Medium Term CAP-0077130	iii. Ensure all emergency lights and exit signs remain unobstructed and clearly visible at all times.
Medium Term CAP-0077125	iv. Install external emergency luminaires to enable safe evacuation during hours of darkness, all the way to a safe place outside the building(s).
Longer Term CAP-0077126	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.36 - Emergency light and exit sign found obstructed in the knitting hall, Value Stream Unit-1



Figure 2.37 - Emergency light not provided in the RO plant room of Colony OHWT ancillary



Figure 2.38 - External emergency lights not provided in the emergency stairs Hosiery Division-5 Canteen building



Figure 2.39 - Insufficient emergency lights and exit sign coverage in internal stairs of Male Non-Executive Hostel



Figure 2.40 - Insufficient emergency light on the backside of LT panels in Hosiery Division 5



Figure 2.41 - Emergency light was found too dim to provide adequate illumination in Gas Boiler Hall



Figure 2.42 - Emergency light missing in the Fire Post ancillary

9.2 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs should be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202604-065009

Timeline	Remediation Plan
Medium Term CAP-0077077	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately upon failure of general lighting power as required by the Building Code of Pakistan Fire Safety Provisions 2016. Emergency lighting shall be powered from a source independent of that supplying the normal lighting.
Longer Term CAP-0077078	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Generator, Transformer & Sub-Station Room

10.1 Ventilation and illumination

This section refers to the LT room.

An assessment of the existing ventilation system was conducted, alongside an evaluation of the lighting setup in various parts of the room, to determine their compliance with recommended standards.

It was observed that the LT room 3 in the Dye House building of the Apparel Plant lacked sufficient ventilation.

BS7671 18th Edition + A2 (2022) specifies the need for ventilation to prevent overheating and ensure the safe operation of electrical installations.F-202604-065094

Timeline	Remediation Plan
Medium Term CAP-0077075	Hire a qualified engineer to reassess the ventilation system to ensure it meets safety and performance standards. If the assessment concludes that the current system is inadequate redesign the ventilation layout to enhance the distribution of airflow, ensuring that critical areas around the HT panel and transformer receive adequate ventilation. This may involve repositioning vents or adding ductwork as necessary. Submit the ventilation assessment report and the proposed design to Pakistan Accord for review.
Longer Term CAP-0077076	Establish regular maintenance and testing of the ventilation system within the spaces to ensure adequate performance.



Figure 2.43 - Insufficient cross ventilation in the LT 3 of Dye House building, Apparel Plant



Figure 2.44 - Insufficient cross ventilation in the LT 3 of Dye House building, Apparel Plant

10.2 Transformer arcing horn

This section refers to the transformer(s) serving the facility.

During the inspection of the transformer(s) serving the facility, the condition of the transformer's arcing horn was assessed. An examination was conducted to identify any signs of damage, wear, or improper installation.

Misaligned arcing horns were observed at the transformer installed near the Apparel gate office. Improper arc diversion may lead to equipment damage and increased risk of fire.

Maintaining the arcing horn in optimal condition is crucial for ensuring life and fire safety. The arcing horn is designed to redirect electrical arcs away from sensitive transformer components, thereby preventing damage and reducing the risk of fire. A worn or misaligned arcing horn may fail to perform this function effectively, increasing the risk of electrical faults, equipment damage, and potential fires. F-202604-065011

Timeline	Remediation Plan
Medium Term CAP-0077073	Realign the arcing horns to the correct gap and position as per manufacturer specifications and relevant standards.
Longer Term CAP-0077074	Establish a routine maintenance schedule for regular inspection and servicing of the arcing horn and related components. Implement measures for early detection of wear and corrosion, such as periodic testing and monitoring, to ensure the arcing horn remains in optimal condition and effectively prevents arcing-related damage.



Figure 2.45 - Misaligned arcing horns observed at transformer installed near Apparel gate office

10.3 Room separation

During the inspection of the generator, transformer, and substation rooms, the physical separation between these critical areas was assessed. The barriers, walls, and partitions intended to isolate each component were reviewed for adequacy and compliance with safety standards.

It was observed that adequate and code-compliant separation was not maintained between oil-type transformers, dry-type transformers, HT panels, and LT panels. In several locations, barriers were either absent or insufficient, with visible gaps and openings that could allow the spread of fire or electrical faults between adjacent sections. Additionally, some partitions were constructed from materials that do not meet required fire-resistance ratings.

This condition was noted in the substation rooms at the Boiler House, Garments Block, Value Stream Unit-1, Colony Workshops, and the LT Room & OHWT ancillary.

The lack of proper segregation significantly increases the risk of fault escalation, fire propagation, equipment damage, and potential injury to personnel. While dry-type transformers may be installed within electrical rooms under controlled conditions, oil-type transformers require proper fire-rated separation, which was found to be inadequate at the time of inspection.F-202604-065093

Timeline	Remediation Plan
Medium Term CAP-0077127	Provide immediate fire risk mitigation by installing temporary fire-resistant barriers to segregate oil-type transformers from HT/LT panels and other equipment. Seal all gaps and openings, and ensure safe clearances are maintained to minimize the risk of fire and fault propagation.
Longer Term CAP-0077128	Redesign and upgrade the substation layout to ensure full compliance with applicable standards (e.g., BS 7671, NFPA guidelines), including dedicated fire-rated compartments for oil-type transformers, appropriate segregation between HT and LT panels, controlled placement of dry-type transformers with adequate clearance, ventilation, and fire-resistant separation, use of compliant fire-rated construction materials with no unprotected openings, and implementation of a periodic inspection and maintenance program to ensure long-term integrity of all segregation measures.



Figure 2.46 - Oil-type transformer and LT panel housed in the same room of Boiler House, violating separation requirements



Figure 2.48 - Oil-type transformers installed in close proximity to each other in Boiler House



Figure 2.47 - Dry-type transformer, HT panels, and LT panels housed in the same room of Garments Block, violating separation requirements



Figure 2.49 - Dry-type transformer, HT panels, and LT panels housed in the same room of Value Stream Unit-1



Figure 2.50 - Oil-type transformer, HT panel, and LT panels housed in the same room of LT room & OHWT ancillary

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

The following observations were noted:

i. It was observed that the cables along the wall near the Fire Hydrant Pump Room OHWT, in the kitchen of Value Stream Unit 2, in the knitting hall on the ground floor of Garments Block, have inadequate support. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure.

ii. Cable trays were found rusted and overcrowded in the RO plant area, Hosiery Division 5 Utility 2. Deteriorated and overcrowded cable trays can damage cable insulation, restrict heat dissipation, and increase the likelihood of short circuits, electrical faults, or fire incidents.

iii. Cables were found laid in an improper and unorganized manner in the cable trenches at the Solar Plant Parking, which can lead to mechanical damage, poor heat dissipation, and increased risk of electrical faults or fire.

iv. 11 kV bare conductors were found routed within the factory premises without mechanical protection or guarding, increasing the risk of accidental contact or damage. This issue was observed specifically near the Colony gate office. F-202604-064996

Timeline	Remediation Plan
Medium Term CAP-0077553	i. Secure cables with appropriate supports to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).
Medium Term CAP-0077554	ii. Replace corroded trays and reorganize cables to maintain proper spacing and load within tray capacity.

Medium Term CAP-0077555	iii. Arrange and secure cables properly within the trenches using suitable supports to ensure segregation, protection, and adequate spacing.
Longer Term CAP-0077556	iv. Install appropriate mechanical protection (e.g., cradle guards or barrier enclosures) for all bare high-voltage conductors in accessible or exposed areas to prevent physical contact and ensure compliance with BS EN 61936-1 and local safety standards.
Longer Term CAP-0077557	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.51 - Improper cable laying observed in cable trench at Solar Plant Parking area



Figure 2.52 - Bare 11 kV conductors routed near colony gate office without protection guards, creating safety risks



Figure 2.53 - Inadequately cables observed near the Fire Hydrant Pump Room OHWT



Figure 2.54 - Cable trays found overcrowded in the RO plant area, Hosiery Division 5 Utility 2



Figure 2.55 - Cable trays were found rusted and overcrowded in the RO plant area, Hosiery Division 5 Utility 2



Figure 2.56 - Inadequately supported cable observed in the knitting hall, Garments Block



Figure 2.57 - Inadequately supported cable observed in the kitchen of Value Stream Unit 2

11.2 Cable trench and channel cleanliness

This is related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of mud and debris, which poses a serious fire risk. This issue was observed specifically in the cable trench near the Solar Plant Parking and in the transformer room of Boiler House.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimize fire risks associated with accumulated debris.F-202604-064995

Timeline	Remediation Plan
Medium Term CAP-0077079	Clean all cable trunking systems to remove accumulated mud and debris, ensuring that any potential fire hazards are mitigated.
Longer Term CAP-0077080	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of mud and debris, thereby maintaining a safe environment and complying with safety standards.



Figure 2.58 - Cable trench filled with mud and debris found in the transformer room, Boiler House



Figure 2.59 - Cable trench filled with dust and debris found in the transformer room, Boiler House



Figure 2.60 - Cable trench filled with mud and debris found near the Solar Plant Parking

11.3 Splicing of cables

This relates to the whole facility.

During the inspection, the cable installation was visually surveyed throughout the facility.

Cables terminations within panels and switchgears were made without cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed specifically in the CDB16-Passenger lift sub-distribution board-3 on the first floor of Garments Block.

For any future electrical work, the termination of stranded cables must be done with cable lugs, and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202604-065993

Timeline	Remediation Plan
Medium Term CAP-0078570	Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0078571	Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.



Figure 2.61 - Cable terminations without lugs observed in the CDB16-Passenger lift sub-distribution board-3 on the first floor of Garments Block

11.4 Colour coding

This relates to the whole facility.

During the inspection, it was observed that an earth connection bar was used for neutral cables. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires. This issue was observed specifically in the lighting distribution board of Gas Boiler Hall and the colony RO plant distribution board.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202604-065994

Timeline	Remediation Plan
Medium Term CAP-0078573	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0078574	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.62 - Earth connection bar used for neutral cables in the lighting distribution board of Gas Boiler Hall



Figure 2.63 - Earth connection bar used for neutral cables in the colony RO plant distribution board

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.